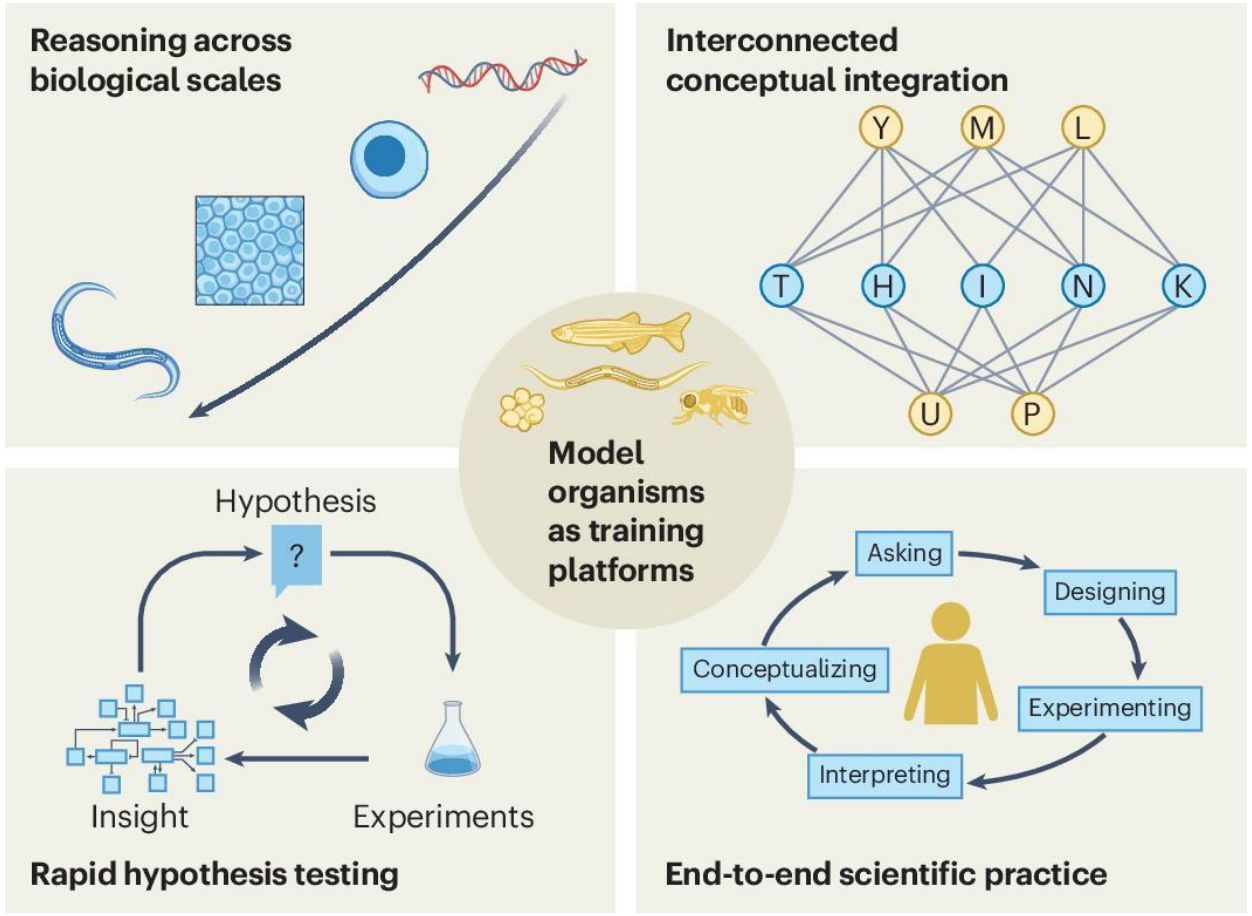
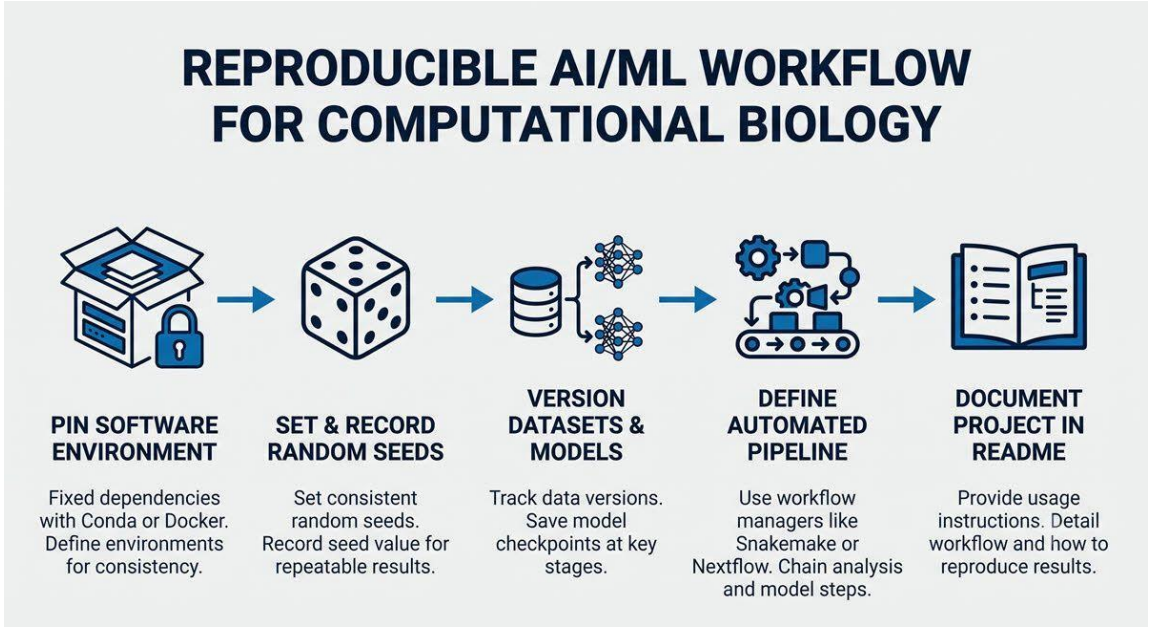
CAN AI GENERATE SCIENTIFICALLY DEFENSIBLE BIOLOGICAL HYPOTHESES FROM GENE EXPRESSION DATA – AND HOW CAN WE EVALUATE THEM?

What did we observe? What can we infer? How can we verify it?

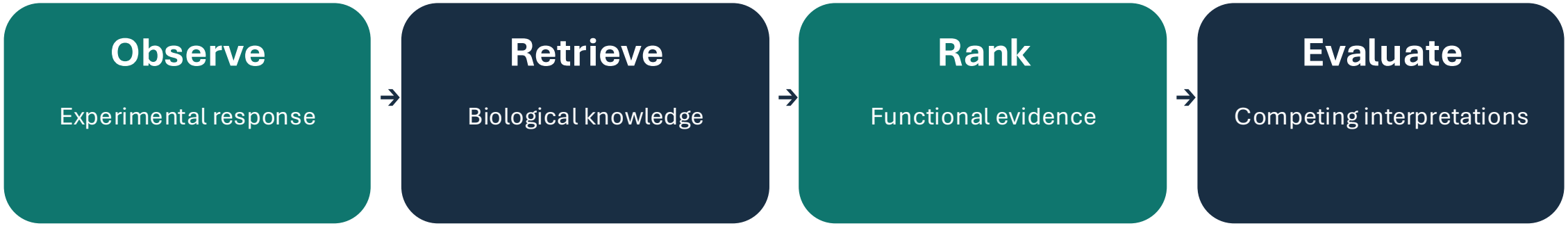
HOW DOES A SCIENTIST MOVE FROM DATA TO A HYPOTHESIS?





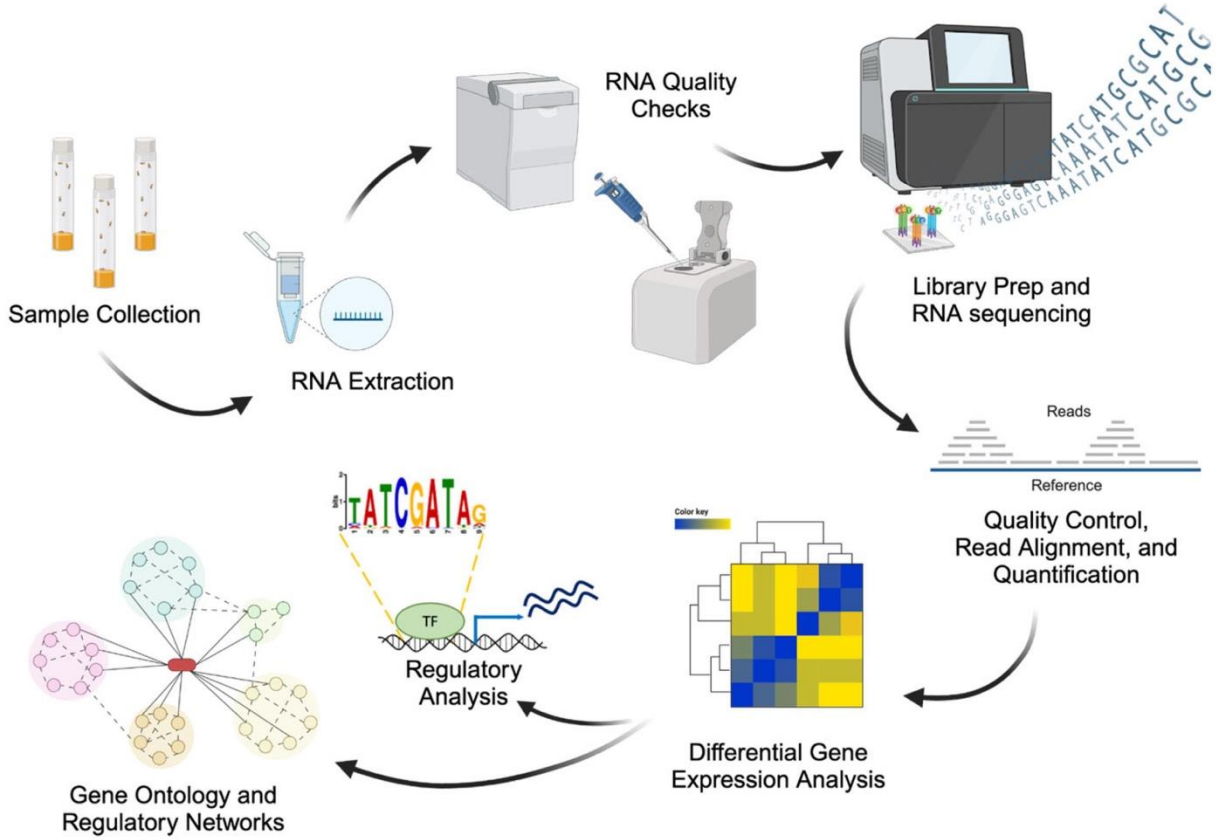
Experiment → Observations → Biological knowledge → Functional patterns → Competing hypotheses → Validation.

HOW DOES A SCIENTIST MOVE FROM DATA TO A HYPOTHESIS?



The computational system formalizes selected steps of a scientist’s reasoning.

WHERE DOES A GENE LIST COME FROM?



Experiment

Control versus treated

→

RNA-seq

Reads and quantification

→

Statistics

DESeq2 / differential expression

→

Input

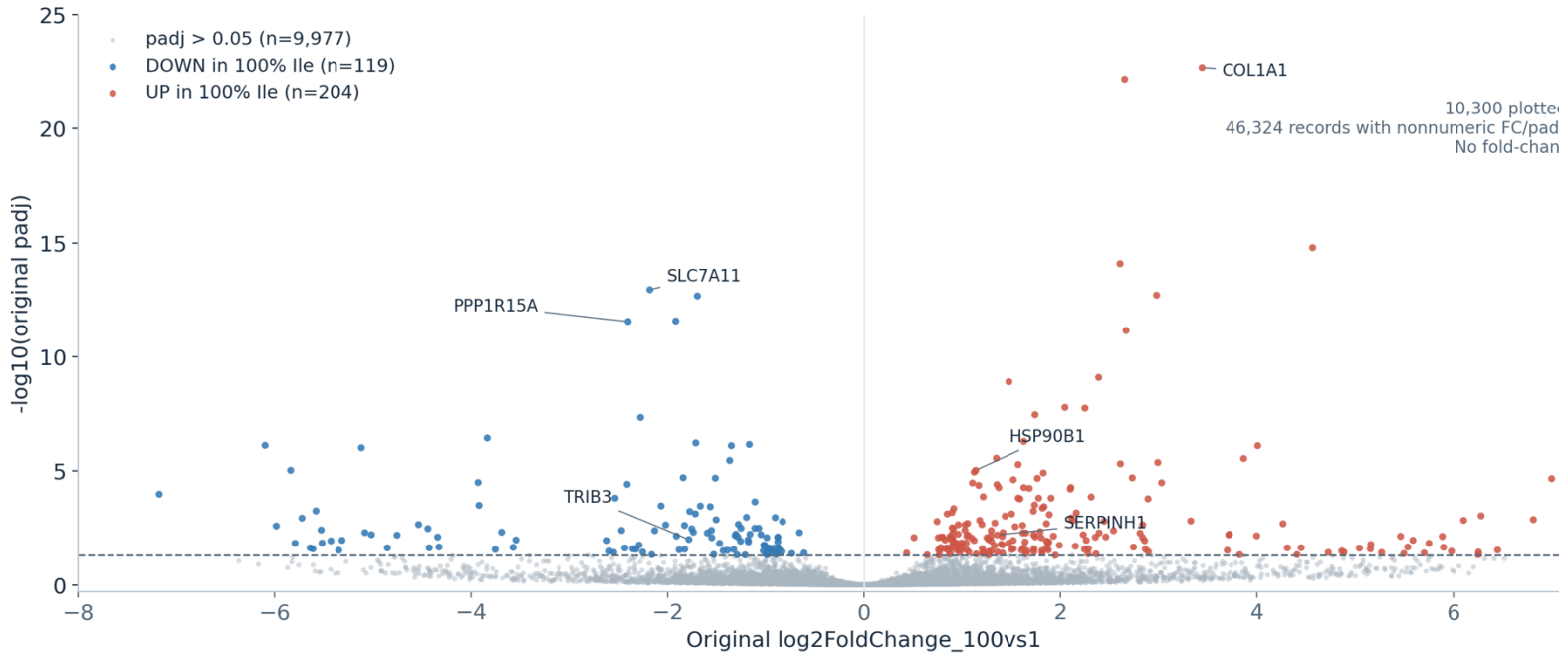
Gene IDs · log2FC · padj

WHAT DO WE ACTUALLY GIVE TO OUR SYSTEM?

PRIMARY RESULT

Isoleucine: original gene-level differential expression

100% versus 1% Ile. Author-provided DESeq2 results; no analysis rerun.



Source run ID(s): 8af2a498af (linked V3 input); original DESeq2 run ID not provided
Source: gkae1184_supplemental_files/Supplementary Table S1 - RNA Sequencing.xlsx | sheet: B_DEGs 100%vs1% Ile
All plot-eligible source rows shown. Missing values were not imputed; selected gene labels are didactic annotations.

FROM GENES TO BIOLOGICAL MEANING

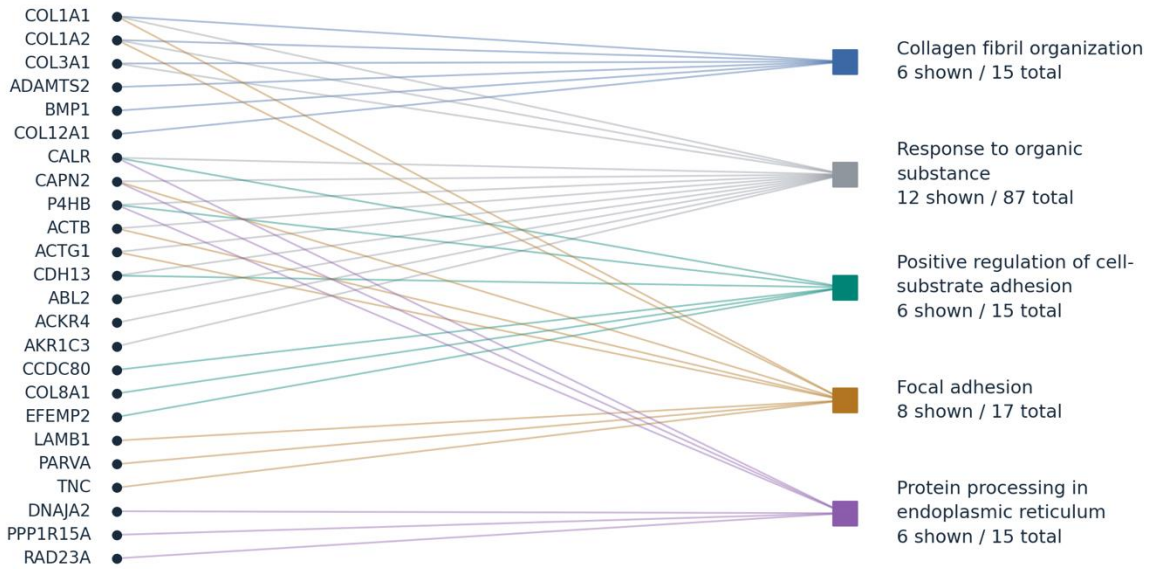
What did we observe? What can we infer? How can we verify it?

A GENE IS NOT A BIOLOGICAL EXPLANATION

DIDACTIC DERIVATION

Observed genes support multiple functional terms

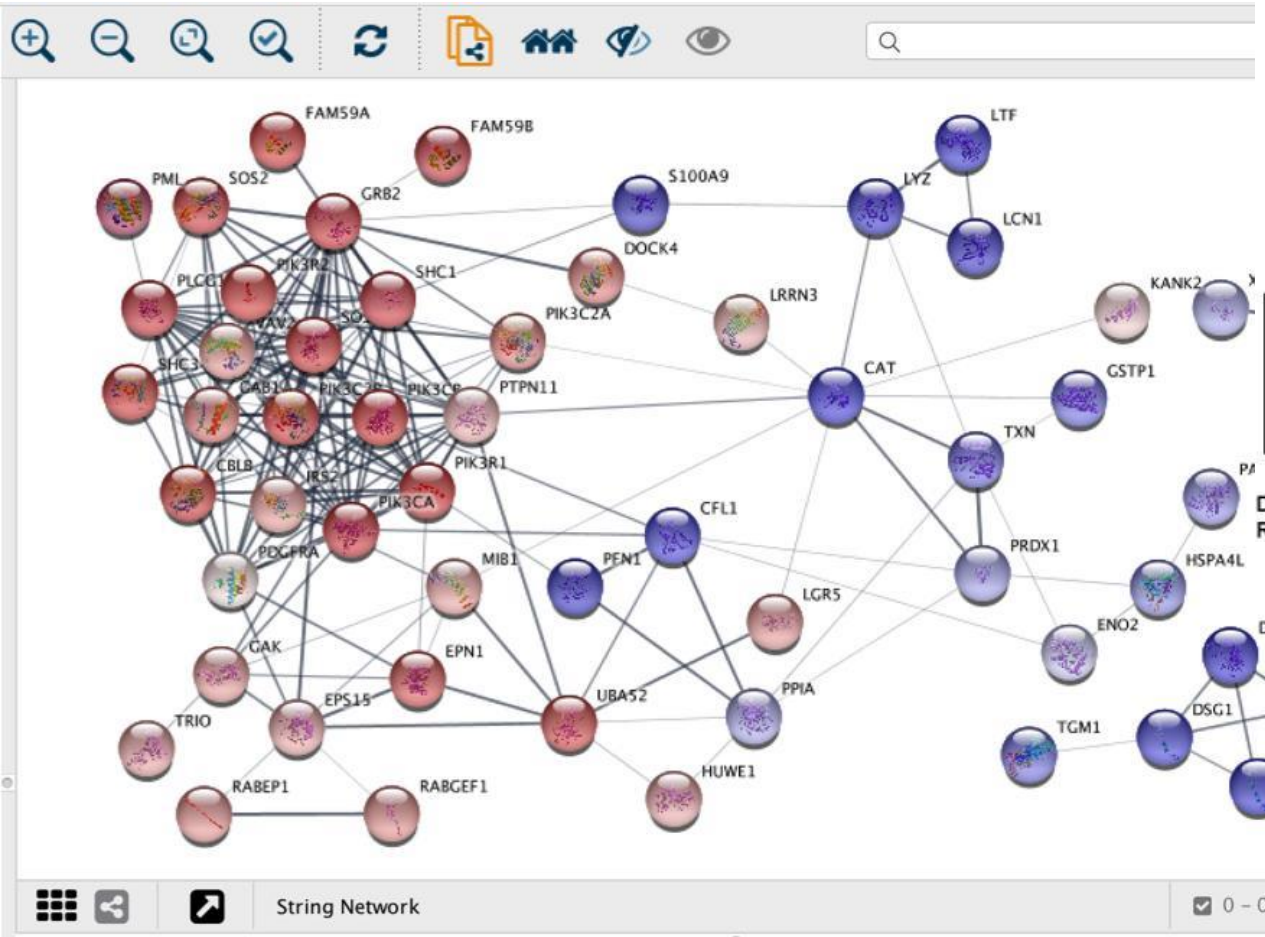
24 representative genes; all five selected V3 terms. Every displayed edge is recorded support.

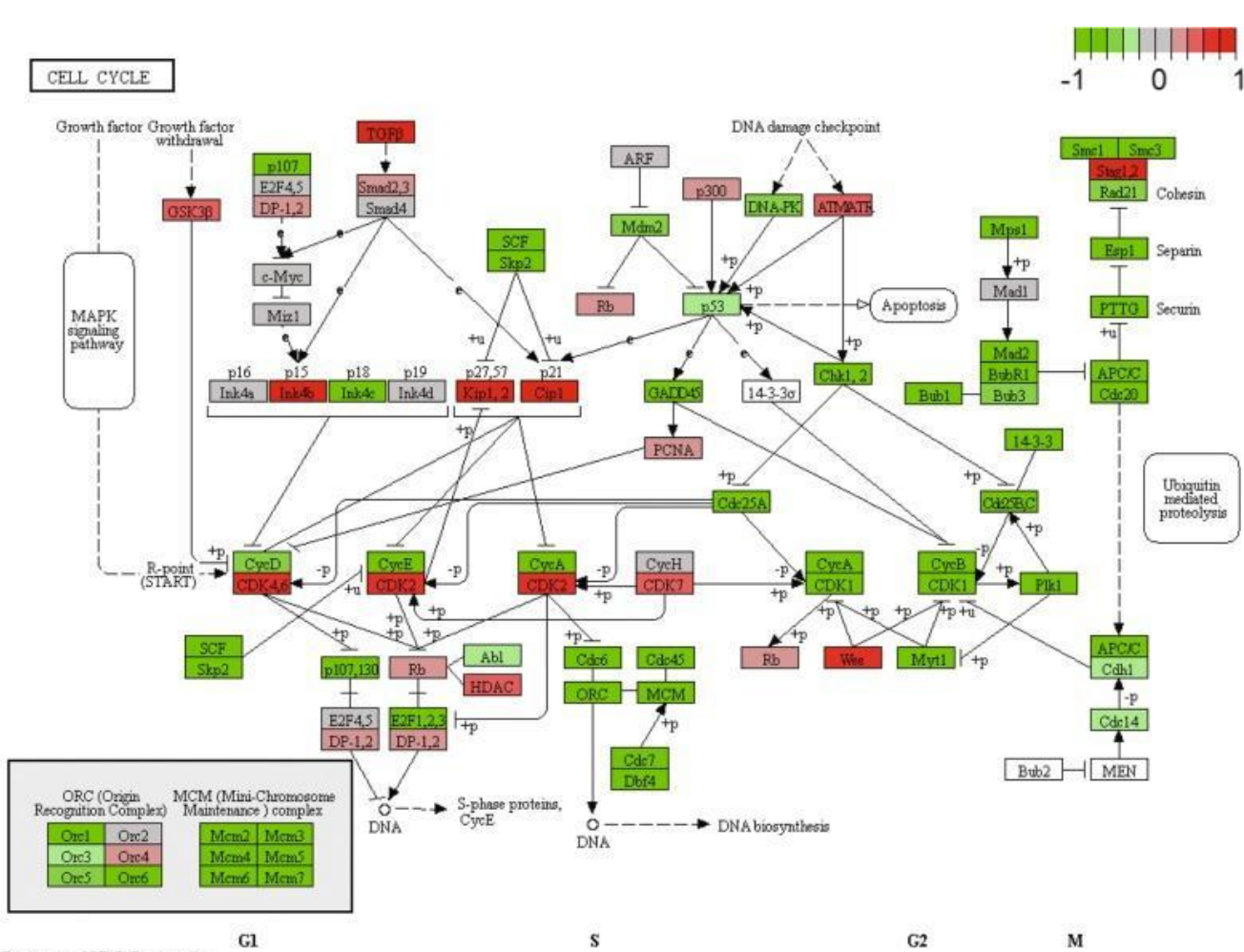


Source run ID(s): 8af2a498af
Source: runs/holdout_2_3_offline_same_raw_replay.json
Display subset: 24/116 supported identifiers. Complete 116-node, 149-edge network is supplied separately. No inferred edges.

STRING, KEGG.

WHERE DO WE LOOK FOR BIOLOGICAL KNOWLEDGE?

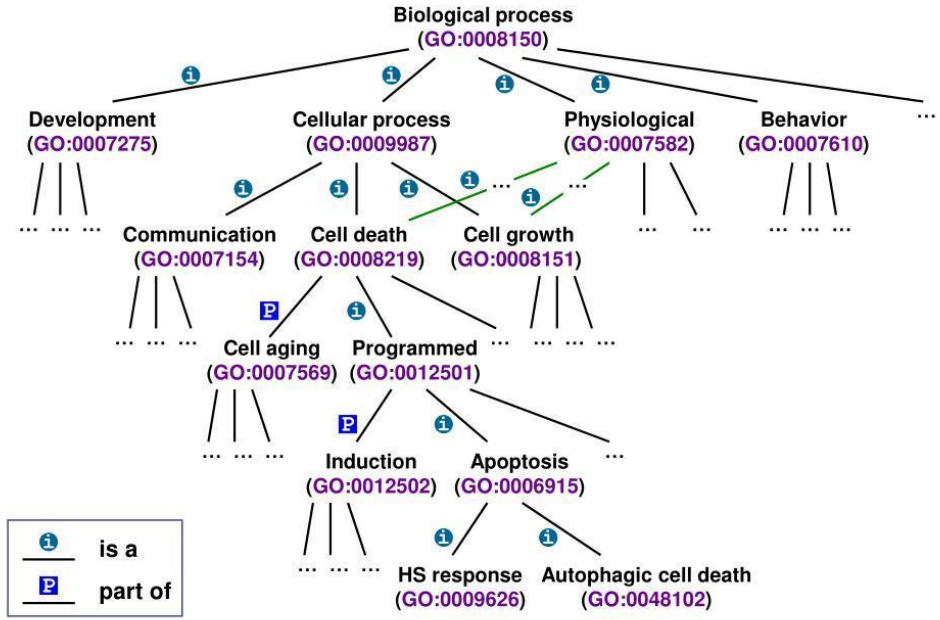


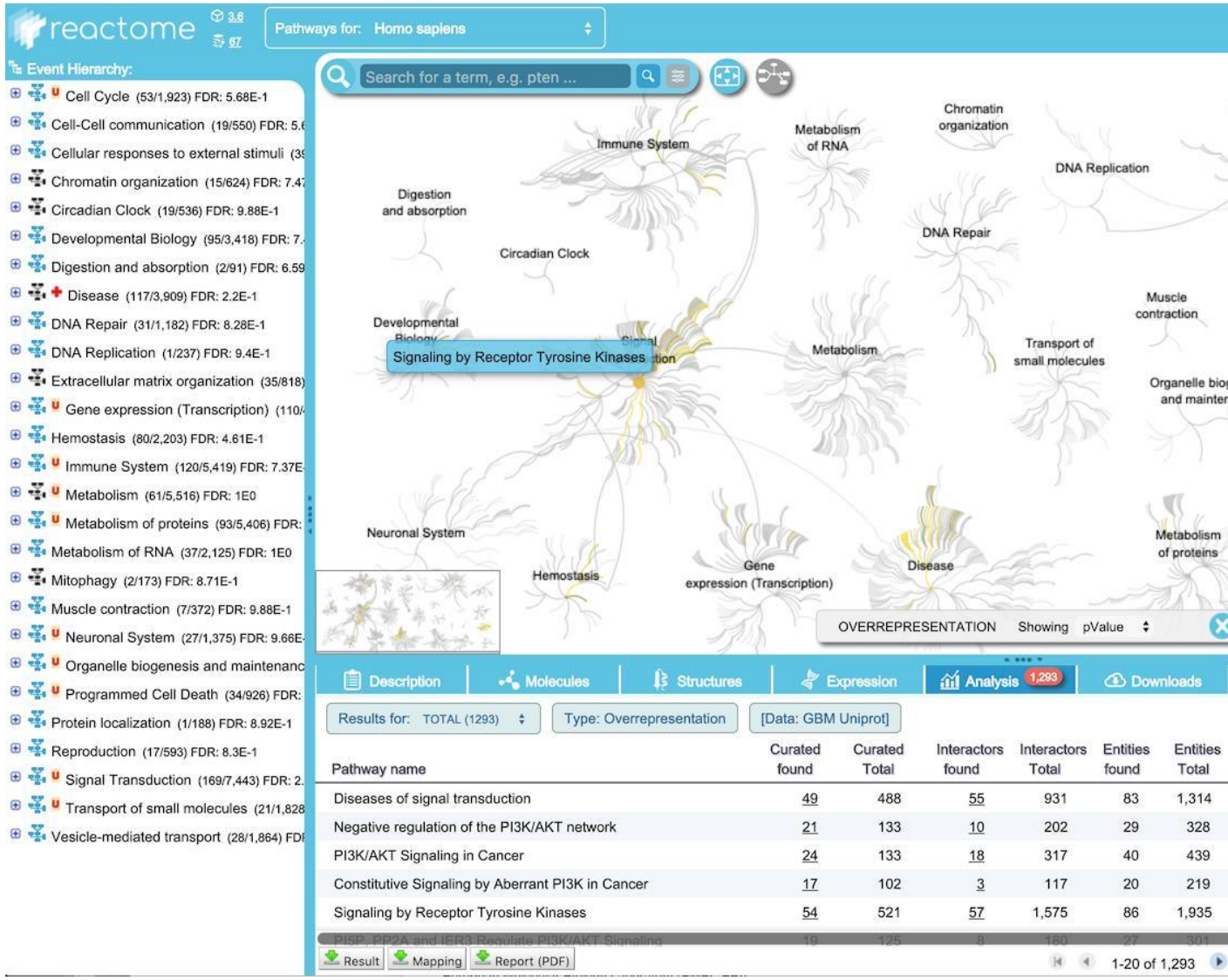


WHERE DO WE LOOK FOR BIOLOGICAL KNOWLEDGE?

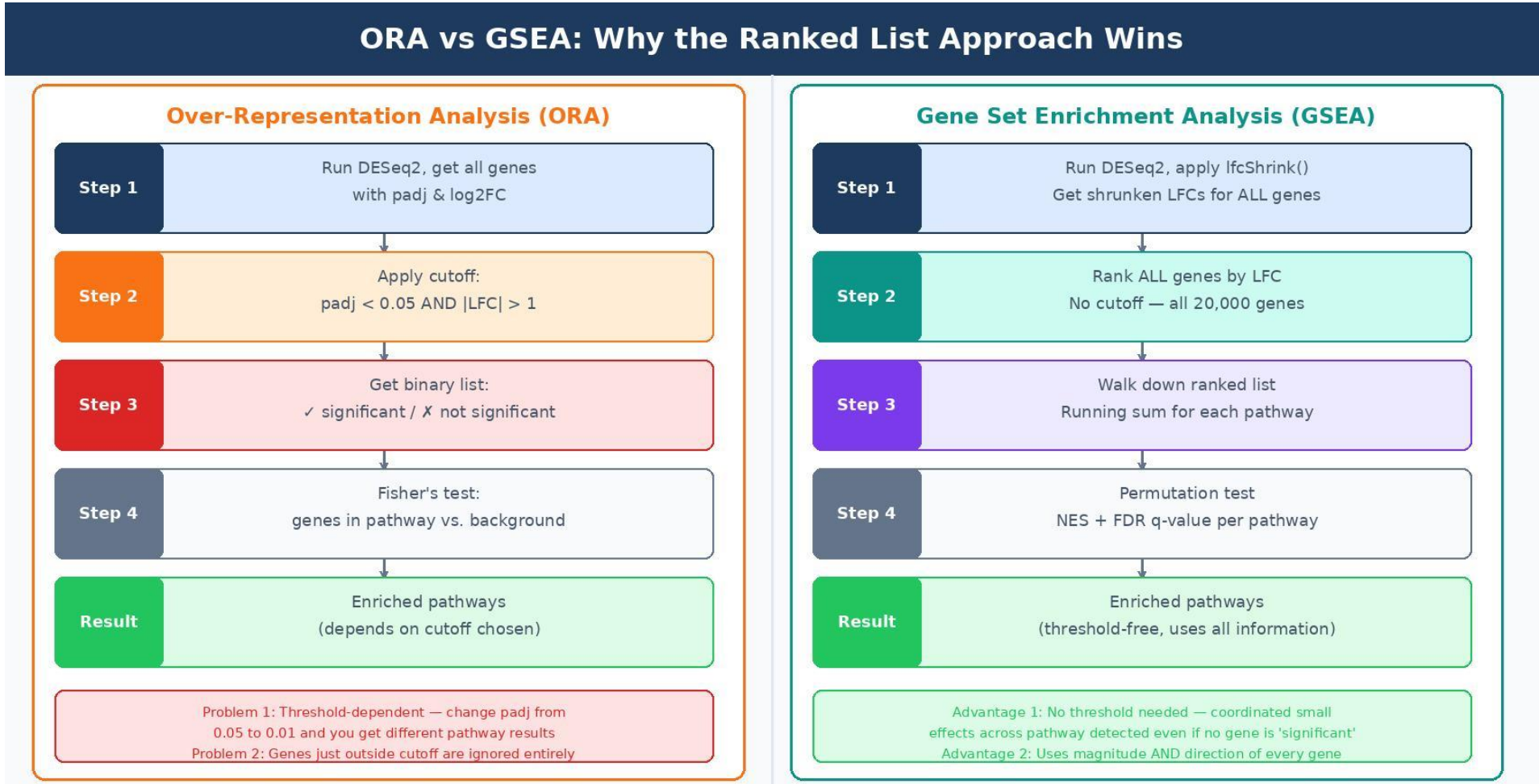
Gene Ontology, Reactome,

Example: Gene Ontology Hierarchy

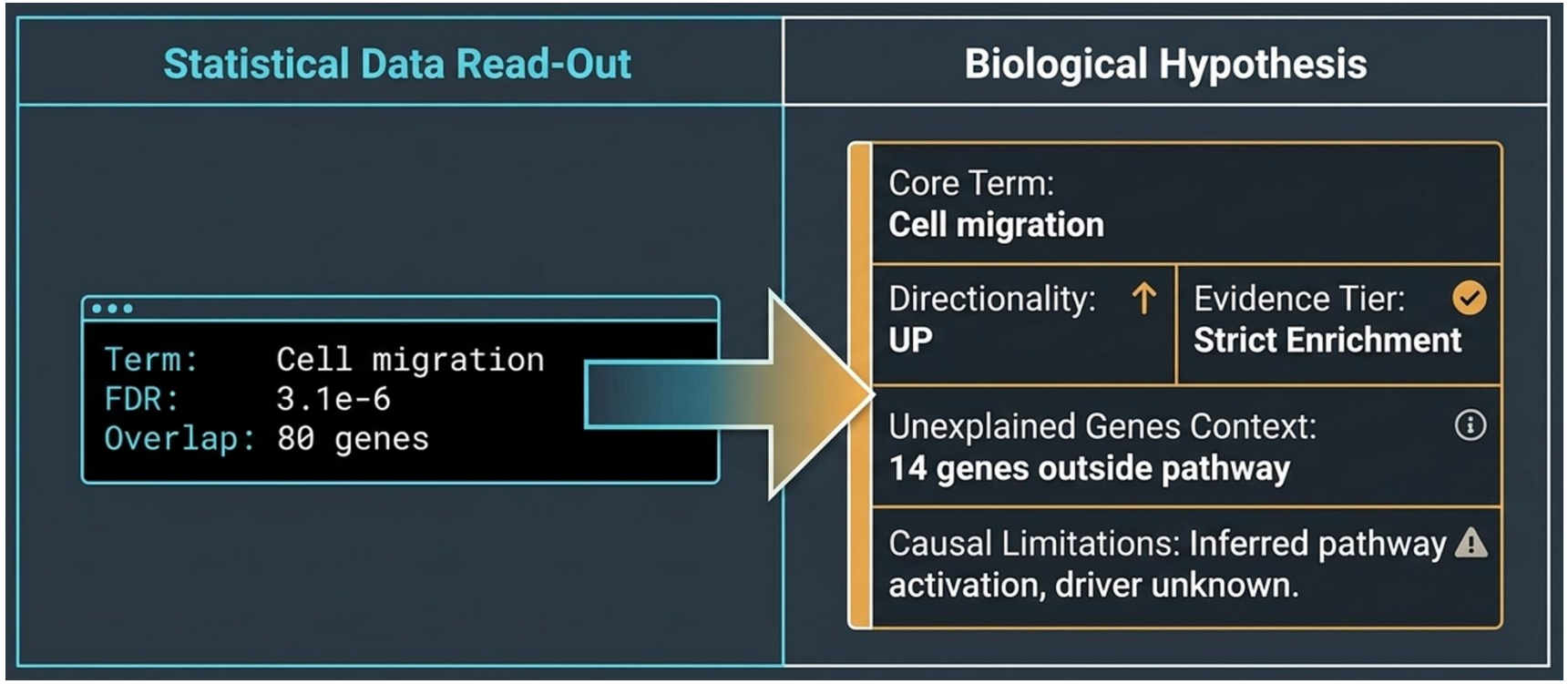




WHAT DOES FUNCTIONAL ENRICHMENT TELL US?

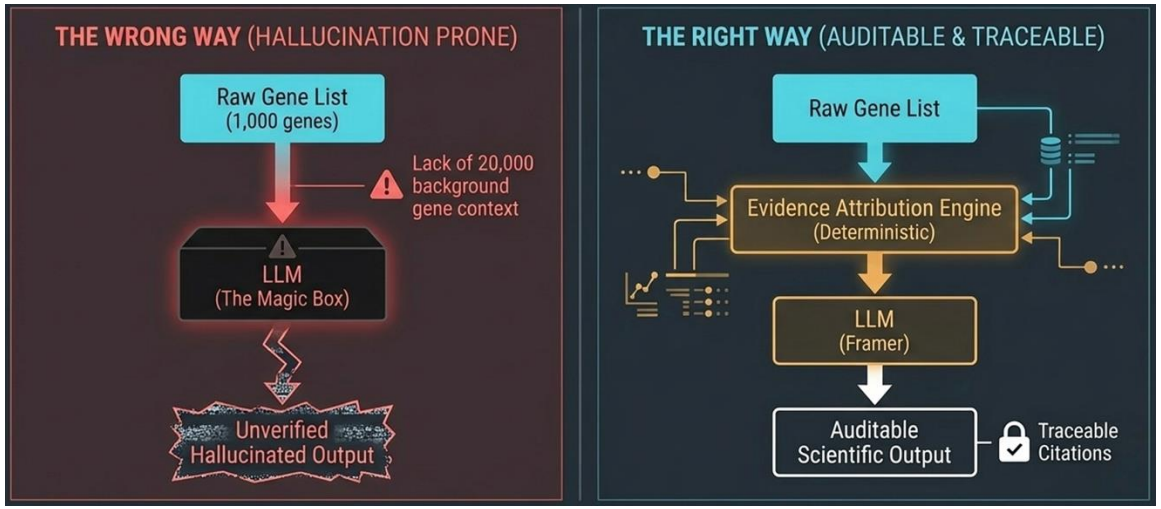


STATISTICAL SIGNIFICANCE IS NOT THE SAME AS INTERPRETATIVE VALUE



It is just a sample

WHY NOT ASK A LARGE LANGUAGE MODEL DIRECTLY?



Gene list

Observations

→

Knowledge

STRING / Reactome

→

Functional V3

Ranked annotations

→

LLM later

Evidence-grounded text

BUILDING GENE HYPOTHESIS MVP

Gene-level input

Identifiers · measured change · source quality

↓

Gene processing + Knowledge retrieval

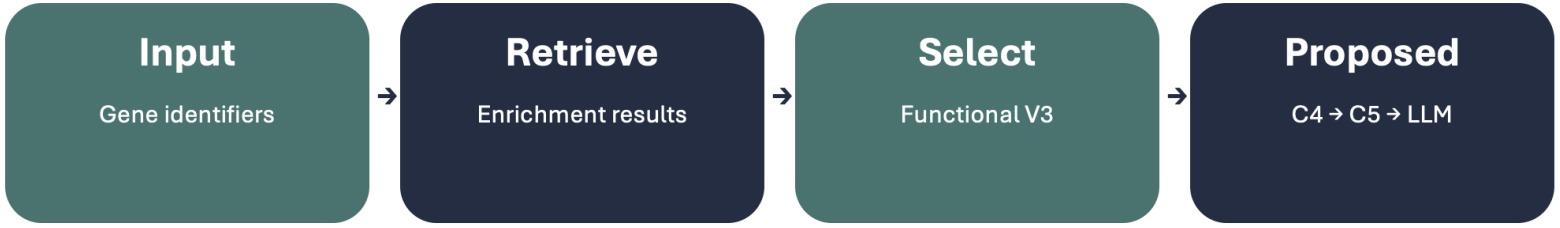
↓

C2 / Functional V3

Enrichment · ranking · diversity

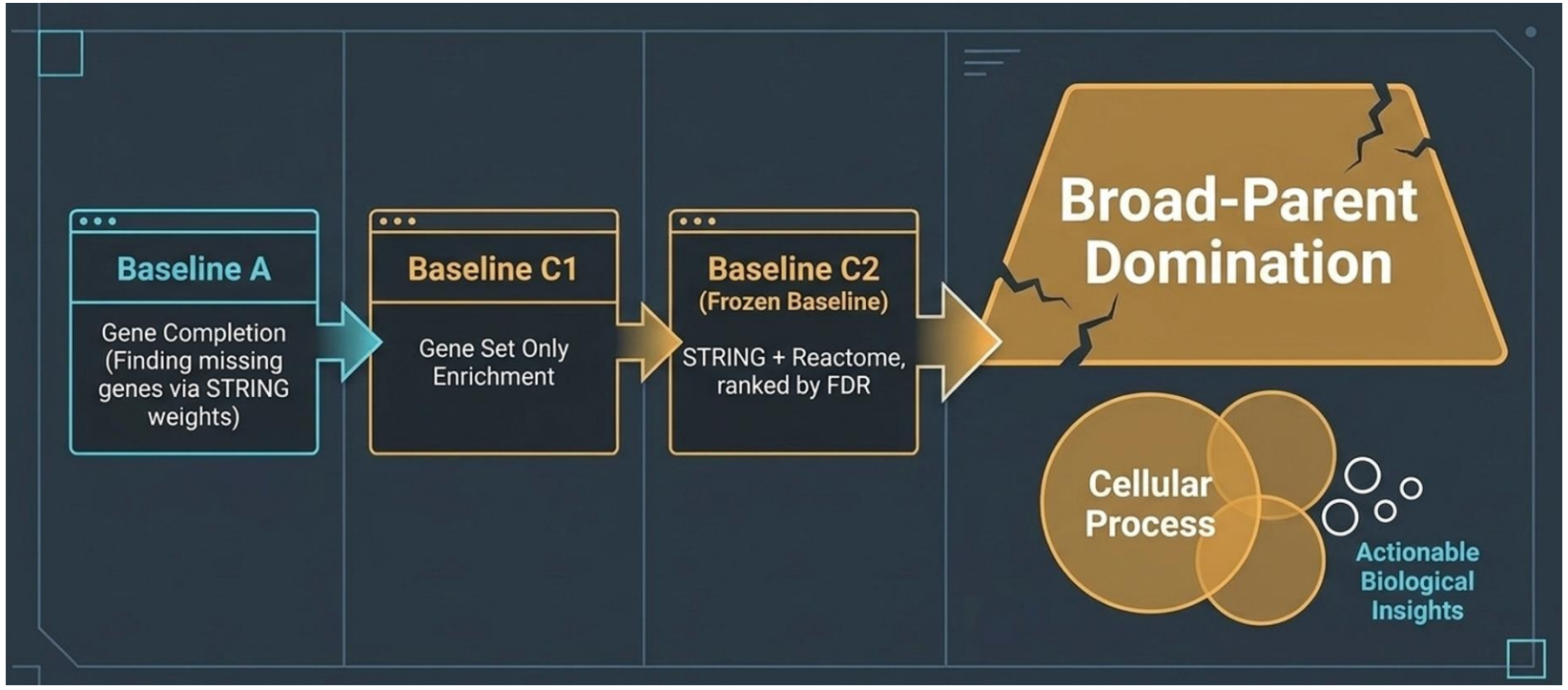
↓

Proposed: Evidence attribution → Structured explanation → LLM



FIRST BASELINES: WHAT WORKED AND WHAT FAIL

Gene completion (A)	Functional interpretation (C1/C2)
Which genes might be missing?	Which biological terms match the genes?
Candidate retrieval and ranking	STRING/Reactome enrichment
Recall@k, MRR	FDR, support, interpretation quality



A REAL EXPERIMENT: 323 DIFFERENTIALLY EXPRESSED GENES

One real experiment: 323 differentially expressed genes

Isoleucine: 100% versus 1% Ile · selected by adjusted p-value ≤ 0.05

323	204	119
ALL submitted identifiers	UP in 100% Ile	DOWN in 100% Ile

Representative IDs from the full observed input:

COL1A1 COL1A2 COL3A1 ADAMTS2 LOX
FN1 ITGA5 VCL RHOA CALR
HSP90B1 P4HB EDEM1 SEL1L
PPP1R15A SLC7A11 GDF15 TRIB3

Displayed subset only. Full 323-gene input was used for V3.

ENRICHMENT GIVES US CANDIDATES, NOT CONCLUSIONS

Enrichment yields candidate terms, not final conclusions

Six selected source-derived records from the Isoleucine ALL saved STRING enrichment

Biological term	Enrichment FDR	Genes
Extracellular matrix organization	2.59e-10	29
Collagen fibril organization	1.90e-09	15
Response to organic substance	3.56e-07	87
Positive regulation of cell-substrate adhesion	7.03e-06	15
Focal adhesion	1.97e-05	17
Protein processing in endoplasmic reticulum	3.14e-05	15

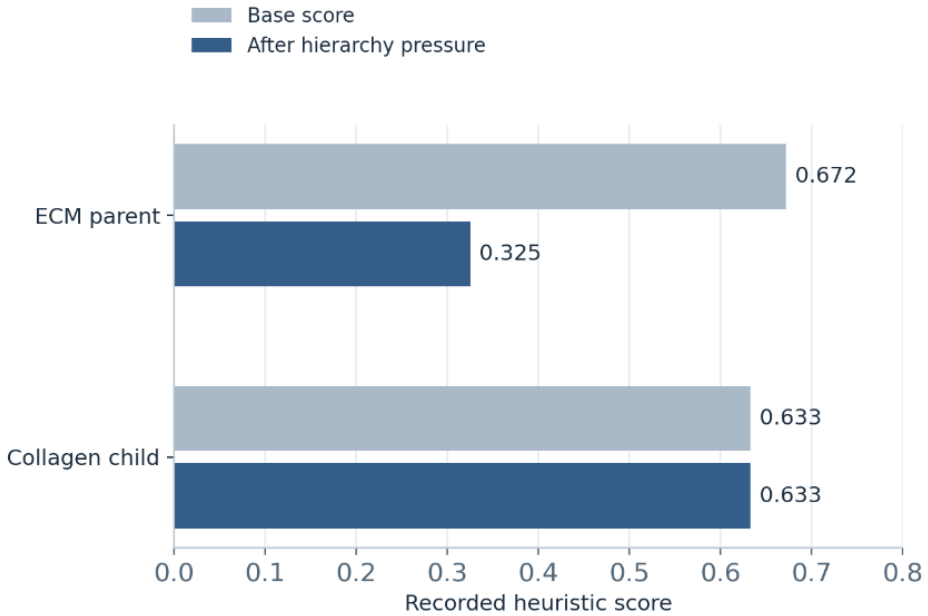
WHY A MORE SIGNIFICANT TERM MAY BE LESS USEFUL

The ECM parent has the lower enrichment FDR; the specific collagen child is selected.

ECM organization
GO:0030198
29 supporting genes; FDR 2.59e-10
Rejected: rank below limit

collagen is_a ECM

Collagen fibril organization
GO:0030199
15 supporting genes; FDR 1.90e-9
Selected: V3 rank 1



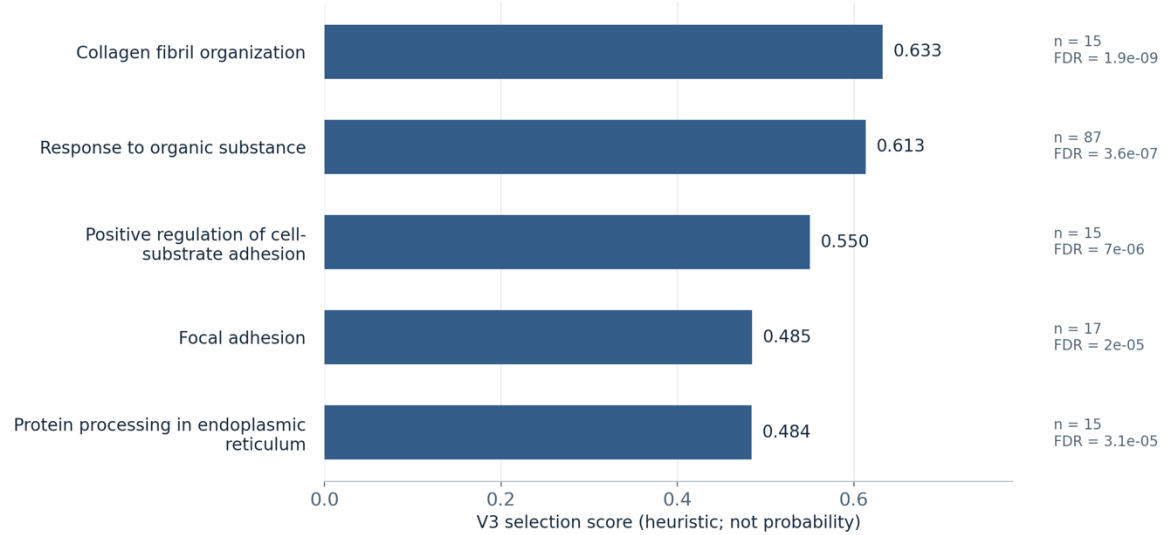
Parent pressure = 0.515568
0.671770 x (1 - 0.515568) = 0.325427

V3 PRODUCES MULTIPLE, PARTIALLY OVERLAPPING INTERPRETATIONS

PRIMARY RESULT

V3 selects five functional interpretations

Isoleucine ALL: 323 submitted identifiers. Scores are recorded SAME-RAW replay results.

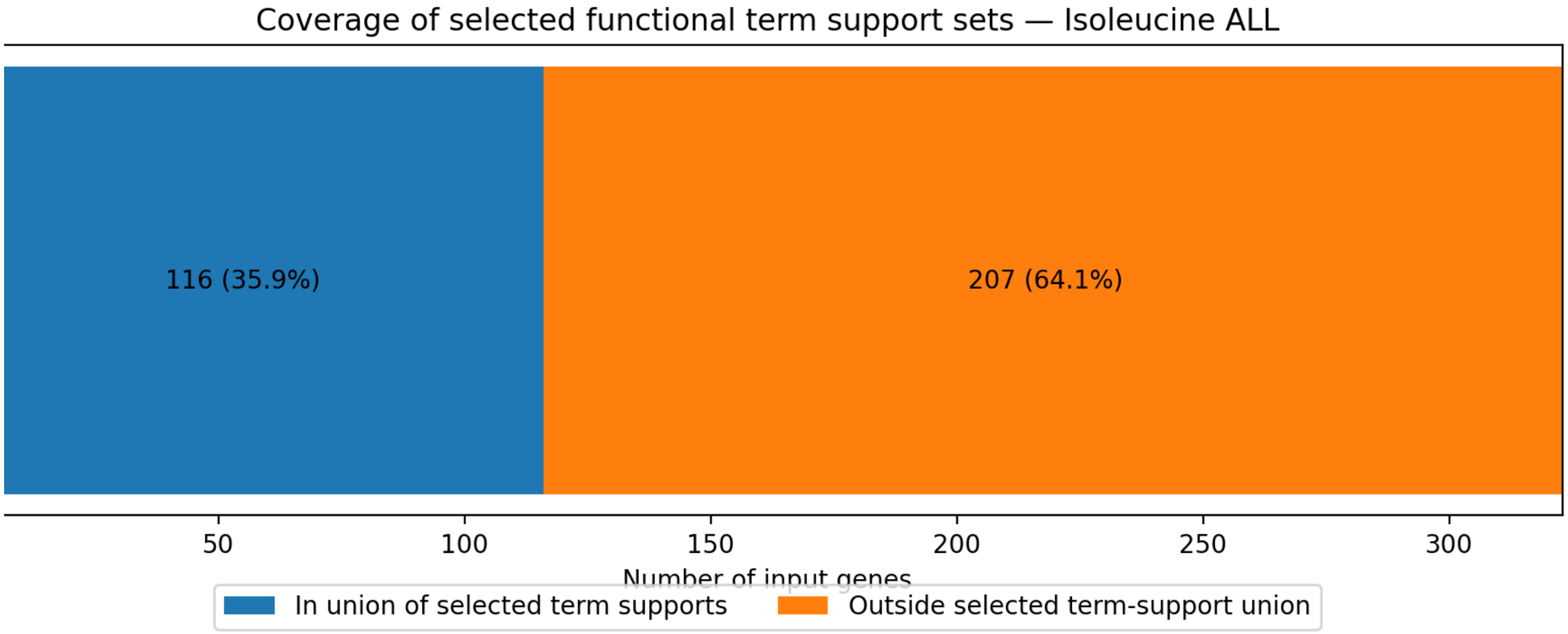


Source run ID(s): 8af2a498af
Source: runs/holdout_2_3_offline_same_raw_replay.json
Displayed rank and scores are unchanged. Five labels do not imply five independent mechanisms.

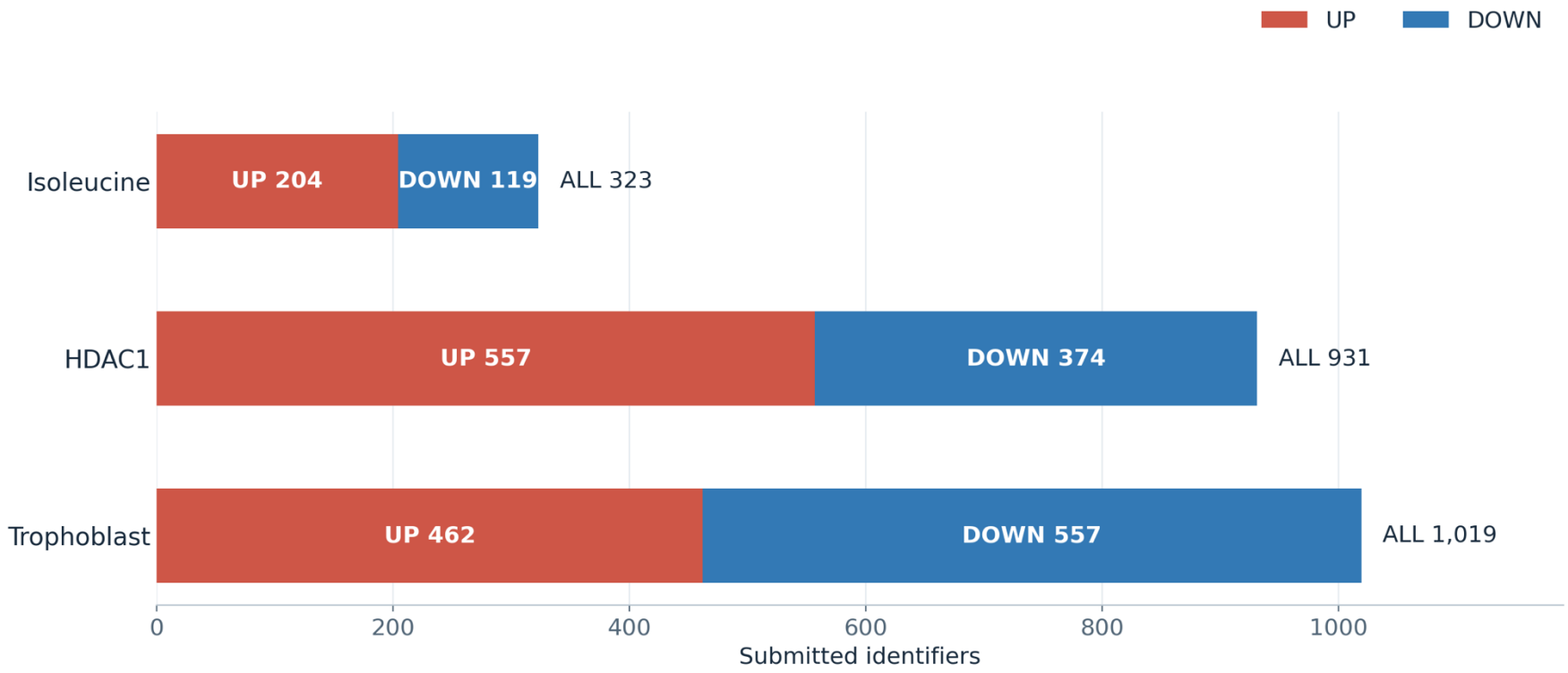
EVIDENCE AND EVALUATION

What did we observe? What can we infer? How can we verify it?

HOW MUCH OF THE INPUT DOES V3 DESCRIBE?



DOES IT WORK ON OTHER BIOLOGICAL EXPERIMENTS?

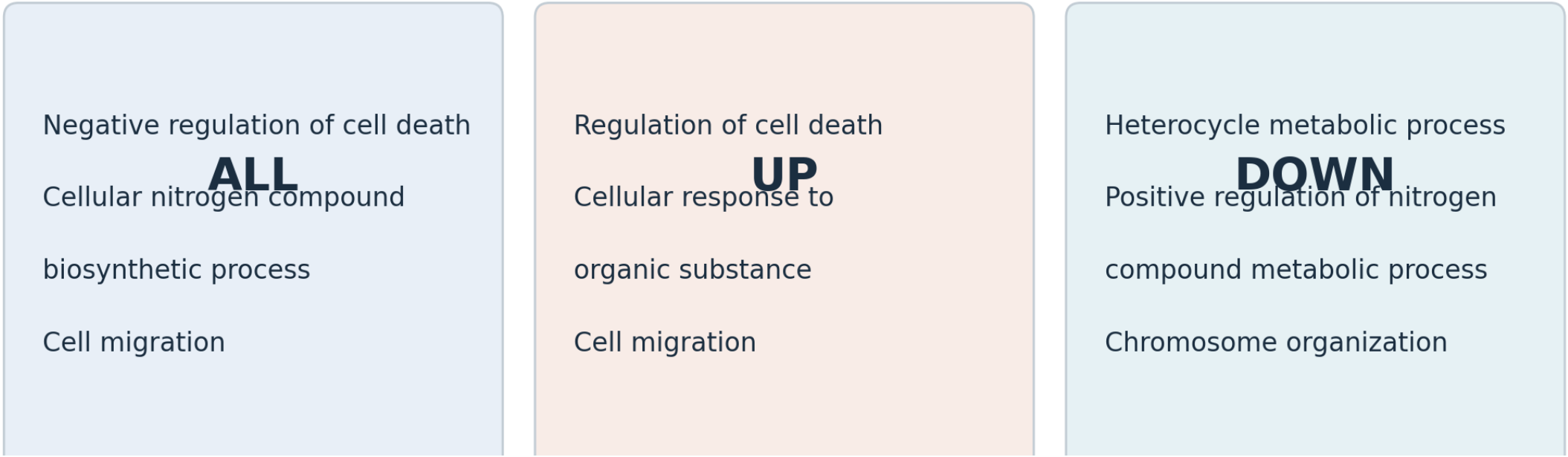


Source run ID(s): Isoleucine 8af2a498af; HDAC1 e201913d84; Trophoblast 9bae950d4d (ALL)

HDAC1: AN INDEPENDENT BIOLOGICAL CONTEXT

HDAC1: functional interpretations across three profiles

Recorded selected terms; labels do not demonstrate biological direction or a causal mechanism



WHAT CAN WE CONCLUDE FROM THE FIRST EVALUATIONS?



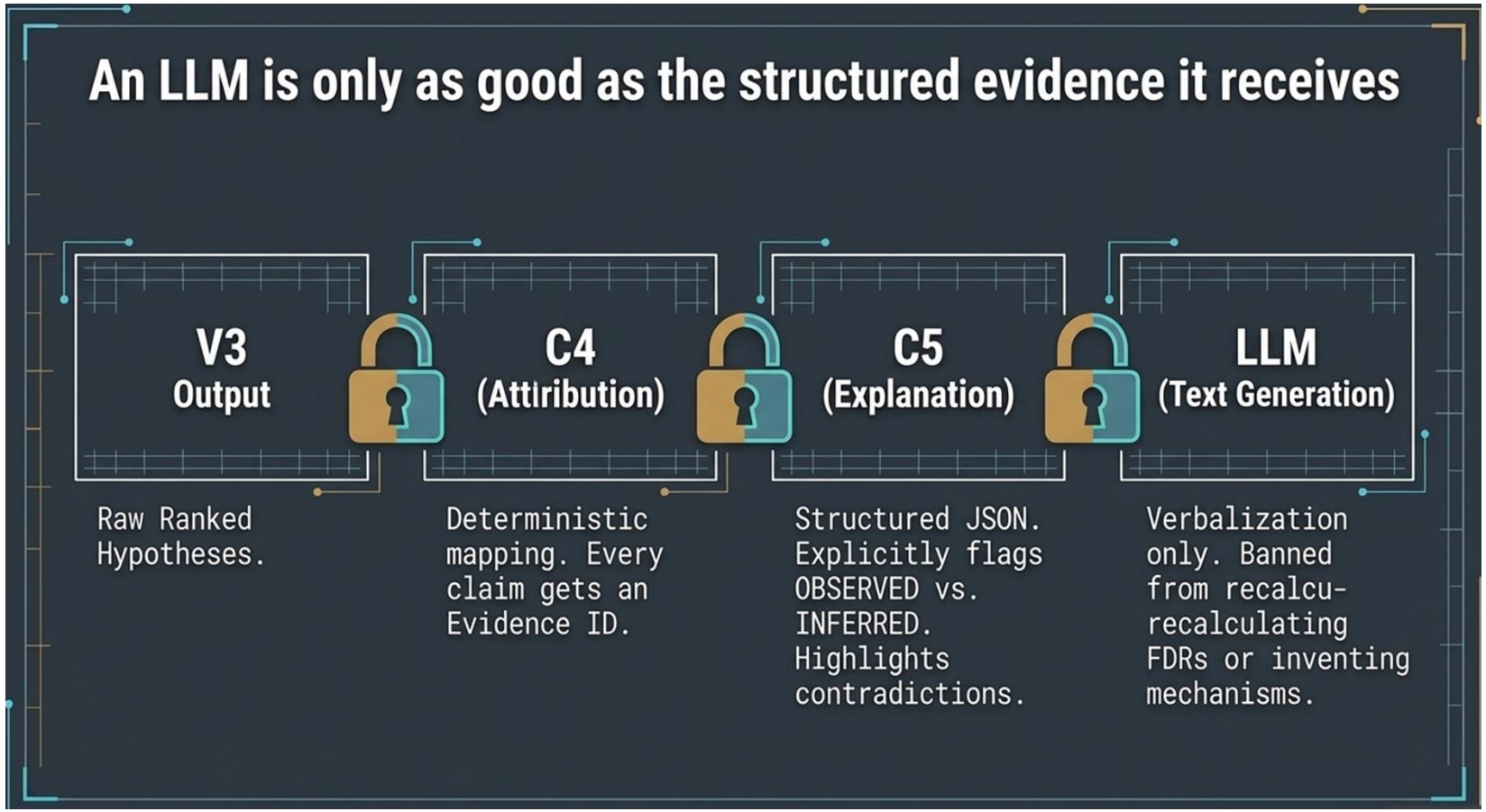
Limitations: dependent ALL/UP/DOWN profiles · no fully documented HDAC1 ratings · scoring rubric not confirmed frozen

DEMONSTRATION AND THE FUTURE PRODUCT

Live Demo: From 323 Genes to Five Interpretations

Step 1 – Inspect the experimental data
Step 2 – Examine the baseline output
Step 3 – Compare with V3 using identical source data
Step 4 – Inspect the ranking decision
Step 5 – Inspect the final interpretations and supporting genes

THE NEXT STEP IS EVIDENCE ATTRIBUTION, NOT ANOTHER ENRICHMENT ALGORITHM



THANK YOU FOR YOUR ATTENTION

Questions & Discussion

Gene Hypothesis MVP

From gene expression to auditable functional interpretations

Current: Functional V3 • Next: Evidence attribution → Structured explanation → LLM